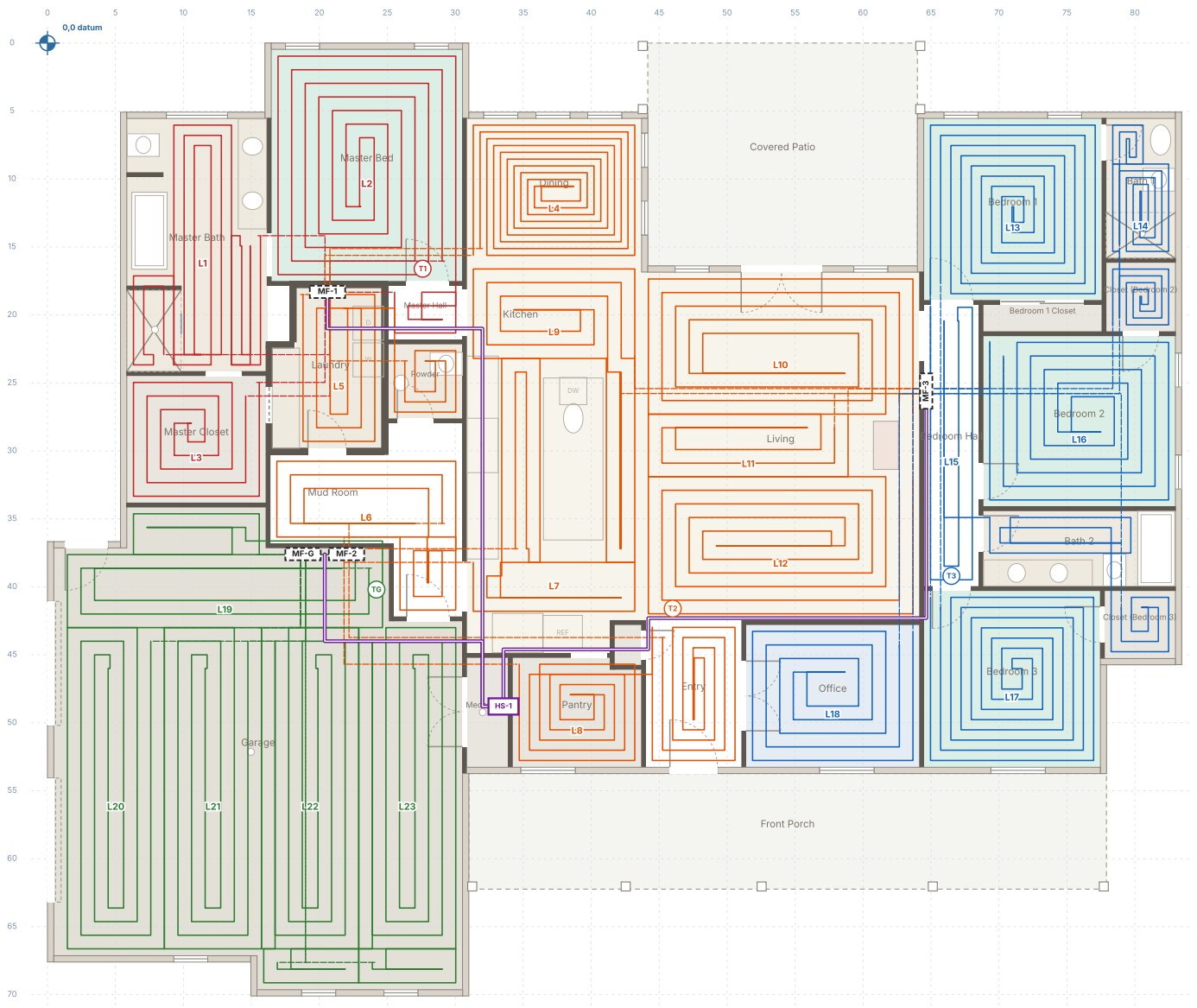
REAR · PATIO



FRONT · PORCH

Zone 1, Master suite (T1) · Zone 2, Great room and utility (T2) · Zone 3, Bedroom wing and office (T3) · Zone G, Garage (held at 50°F) (TG)

leader pair (drawn spread out) · 1" feed pair from HS-1 · manifold

23 loops, 4,920' of 1/2" PEX, 4 manifolds. · Red zone 1, orange zone 2, blue zone 3, green garage. · 54 kBtu/h at design, 110°F supply. · Zones are thermostats on loop actuators, not manifolds.

draft

R-1 Radiant · 64 Stone Sheep Circle

PEX loops in the slab, four thermostat zones, manifolds and flows at Powell, WY design (-11°F). App version v6. From the owner's plans (Larson House, 6/11/26).

Notes

1. Positions are taped from the nearest wall face, exterior walls first. Every panel stops 6" off the walls, so interior wall plates fasten in that band; mark every tube crossing under a wall line on the vapor barrier and keep shots and anchors 12" from it.
2. Tube: 1/2" O2-barrier PEX in counterflow spirals (supply in, return out between), tied to the mesh at mid-depth of a 4" slab. 8 loops run at 9" o.c. or 6" (baths, and rooms needing 15 Btu/h per sq ft or more; 6" in the dining room and Bath 1), the rest at 12". One continuous coil per loop, no fittings in the slab. Tag both ends of every loop at its manifold.
3. No tube under the kitchen base cabinets, island, fridge or back cabinet, the tubs, the vanities, the laundry cabinets or the fireplace hearth. Keep tube 6" from toilet flanges and 12" from floor and shower drains; the showers' floors are heated.
4. Under the slab: R-10 (2") rigid foam under the whole heated slab and the garage slab, with R-10 at the slab edge (IECC 2021 zone 6 for a heated slab), and a 10-mil vapor barrier over the foam.
5. Zoning is by thermostat: four zones (master suite, great room and utility, bedroom wing and office, garage) on 24 V actuators at each loop's manifold port, so a room can get its own thermostat later by rewiring, without new tube.
6. Manifolds, three stations: MF-1 (laundry, on the north wall between the linen alcove and the dryer) serves Master Bed, Master Bath, Master Closet, Master Hall, Laundry, Powder, Dining; MF-2 (garage, on the mud room wall, beside MF-G) serves Mud Room, Kitchen, Pantry, Entry; MF-G (garage, on the mud room wall, beside MF-2) serves Garage; MF-3 (bedroom hall, recessed in the west wall north of the fireplace) serves Kitchen, Living, Office, Bedroom Hall, Bedroom 1, Bath 1, Closet (Bedroom 2), Bedroom 2, Bath 2, Bedroom 3, Closet (Bedroom 3). HS-1 in the mechanical closet feeds all three.
7. Leaders run in the slab as tight pairs (drawn spread out). Sleeve them in 3/8" foam where they pass under another zone (136' in all) and wherever they cross a sawcut control joint. Feed pairs from HS-1 to

Open items

- L14 (Closet (Bedroom 2), Bath 1): the floor carries all but about 135 Btu/h of the room at -11°F with 110°F water. Add a small electric towel warmer, or run the boiler hotter on the coldest days.
- Heat source: condensing boiler (recommended; it can also make the domestic hot water and replace P-1's 50 gal heater), an air-to-water heat pump, or an electric boiler. HS-1 is drawn as a natural gas boiler, wall-hung on the mechanical closet's east wall, matching E-1's gas assumption; all-electric changes E-1's service size.
- Garage: kept at 50°F on its own manifold, pump and thermostat. Decide whether it runs a glycol mix behind a plate heat exchanger, which protects it if the house loses power in a cold snap with the doors open.
- Flooring: the loops assume hard-surface floors. Carpet in a bedroom needs about 10°F warmer water for that loop, or 9" spacing there.
- Mechanical closet: HS-1, pump P-R1 and the zone controller share it with P-1's water manifold and water heater; that is why no manifold is drawn there (O5 in the options shows the fourth manifold saves little tube). A combi boiler removes the water heater and makes room.
- MF-3 cabinet depth in the hall's 2×4 west wall (S-1), and the boiler, pump and zone controller circuits (E-1).

the three stations are 1" PEX, insulated end to end, 122' each way in all.

8. Flow: 10.9 gpm with every loop calling, each loop set on its flow meter at a 10°F drop. Design heat 54,493 Btu/h, of which 10,834 is the garage at 50°F; 10% is added for heat lost down and out the edge.
9. Water: 110°F supply at -11°F outside, reset lower as it warms, so any low-temperature source can run it. Floor surfaces stay at or under 90°F. At that supply L14 (Closet (Bedroom 2), Bath 1) falls about 135 Btu/h short at design; see the open items.
10. Pump: the hardest loop is L4 at 8.0 ft in the tube; add about 3 ft for the feed pair, manifold and actuator. A constant-ΔP ECM circulator at about 12 ft covers it.
11. Pressure test every loop at 100 psi water or 60 psi air for 24 hours before the pour and keep it under pressure through the pour. Photograph the layout with a tape in frame before concrete.

Loop schedule

Leader is each way. Heat includes the share lost downward. Pick a row to find the loop.

Loop	Zone	Rooms	Manifold	Spacing	In slab	Leader	Total	Btu/h	Flow	Head	Floor
L1	1	Master Bath	MF-1	9"	196'	15'	226'	3,800	0.76 gpm	5.5 ft	85°F
L2	1	Master Hall, Master Bed	MF-1	12"	264'	10'	284'	3,182	0.64 gpm	5.1 ft	76°F
L3	1	Master Closet	MF-1	12"	94'	15'	124'	1,499	0.30 gpm	0.6 ft	79°F
L4	2	Dining	MF-1	6"	248'	16'	280'	4,149	0.83 gpm	8.0 ft	87°F
L5	2	Laundry, Powder	MF-1	12", 9"	114'	8'	131'	1,159	0.23 gpm	0.4 ft	77°F
L6	2	Mud Room	MF-2	12"	123'	9'	142'	1,360	0.27 gpm	0.6 ft	76°F
L7	2	Kitchen	MF-2	12"	116'	13'	142'	1,097	0.22 gpm	0.4 ft	76°F
L8	2	Pantry, Entry	MF-2	9"	205'	27'	260'	2,930	0.59 gpm	4.0 ft	81°F
L9	2	Kitchen	MF-3	12"	160'	26'	212'	1,412	0.28 gpm	0.9 ft	76°F
L10	2	Living	MF-3	12"	212'	5'	221'	2,760	0.55 gpm	3.1 ft	76°F
L11	2	Living	MF-3	12"	90'	10'	111'	963	0.20 gpm	0.3 ft	76°F
L12	2	Living	MF-3	12"	232'	10'	253'	2,796	0.56 gpm	3.6 ft	76°F
L13	3	Bedroom 1	MF-3	9"	225'	10'	246'	3,766	0.75 gpm	5.9 ft	81°F
L14	3	Closet (Bedroom 2), Bath 1	MF-3	6"	137'	27'	191'	2,161	0.43 gpm	1.7 ft	90°F
L15	3	Bedroom Hall, Bath 2, Closet (Bedroom 3)	MF-3	12", 9"	188'	20'	229'	2,371	0.47 gpm	2.5 ft	80°F

Loop	Zone	Rooms	Manifold	Spacing	In slab	Leader	Total	Btu/h	Flow	Head	Floor
L16	3	Bedroom 2	MF-3	12"	175'	10'	195'	2,599	0.52 gpm	2.5 ft	78°F
L17	3	Bedroom 3	MF-3	9"	214'	18'	251'	3,480	0.70 gpm	5.3 ft	81°F
L18	3	Office	MF-3	12"	131'	22'	174'	2,175	0.43 gpm	1.6 ft	79°F
L19	G	Garage	MF-G	12"	195'	7'	209'	1,912	0.38 gpm	1.5 ft	56°F
L20	G	Garage	MF-G	12"	196'	19'	233'	2,118	0.42 gpm	2.1 ft	56°F
L21	G	Garage	MF-G	12"	196'	11'	219'	2,121	0.42 gpm	1.9 ft	56°F
L22	G	Garage	MF-G	12"	248'	22'	292'	2,341	0.47 gpm	3.1 ft	56°F
L23	G	Garage	MF-G	12"	248'	24'	297'	2,342	0.47 gpm	3.1 ft	56°F
Zone 1	1	Master suite	MF-1		3 loops		633'	8,482	1.7 gpm		
Zone 2	2	Great room and utility	MF-1, MF-2, MF-3		9 loops		1,751'	18,626	3.7 gpm		
Zone 3	3	Bedroom wing and office	MF-3		6 loops		1,286'	16,551	3.3 gpm		
Zone G	G	Garage (held at 50°F)	MF-G		5 loops		1,250'	10,834	2.2 gpm		

Manifolds

Tag	Loops	Where	Zones	Feed
MF-1	5 loops: L1, L2, L3, L4, L5	Laundry, on the north wall between the linen alcove and the dryer	1, 2	1" pair F-1, 43' each way
MF-2	3 loops: L6, L7, L8	Garage, on the mud room wall, beside MF-G	2	1" pair F-2, 24' each way
MF-G	5 loops: L19, L20, L21, L22, L23	Garage, on the mud room wall, beside MF-2	G	1" pair F-2, 24' each way
MF-3	10 loops: L9, L10, L11, L12, L13, L14, L15, L16, L17, L18	Bedroom hall, recessed in the west wall north of the fireplace	2, 3	1" pair F-3, 54' each way

Manifold options

The same loops routed to each layout, every loop to its nearest station (two 12-port manifolds at most per station). Leaders count both ends and the 2' rise at the manifold. ★ is drawn.

Option	Layout	1/2" PEX	Leaders	Longest / shortest loop	Loops over 300'	Leader under another zone	Feed pairs	Stations
O1	All manifolds in the mechanical closet, beside the heat source	5,763'	1,572'	354' / 143'	3	420'	none	1
O2	Owner's first sketch: laundry corner, two on the garage side of the mud room wall, two in the bedroom hall beside the fireplace	4,918'	718'	302' / 108'	1	105'	122' × 2	3
O3	Two stations: the mechanical closet, and the bedroom hall	5,226'	1,029'	334' / 111'	2	257'	54' × 2	2
O4 ★	Three stations: laundry north wall (MF-1), mud room wall in the garage (MF-2 and MF-G), bedroom hall north of the fireplace (MF-3)	4,927'	718'	294' / 111'	0	109'	122' × 2	3
O5	O4 plus a fourth manifold in the mechanical closet for the kitchen, pantry and entry	4,882'	671'	291' / 111'	0	90'	122' × 2	4

Bill of materials

Rough-order quantities from the drawing geometry, for pricing. The installer takes off the final list.

Item	Qty	Notes
1/2" O2-barrier PEX, 1,000' coils	6 coils	4,920' + 5% waste; one length per loop, no splices
1" PEX supply + return, insulated	244 lf	HS-1 to the three manifold stations, under the slab
Stainless manifolds with flow meters	4	MF-1 5-loop, MF-2 3-loop, MF-G 5-loop, MF-3 10-loop
24 V thermal actuators	23	one per loop
Zone controller + thermostats	1 + 4	T1, T2, T3, TG; outdoor reset sensor
Circulators	2	P-R1 constant ΔP for the house, P-RG for the garage with its mixing valve
Manifold cabinets	4	recessed at MF-3, surface at MF-1 (sound wall), MF-2 and MF-G
R-10 rigid foam, 2"	4,167 sf	under the heated slab and the garage, plus the slab edge
10-mil vapor barrier	4,584 sf	over the foam, lapped and taped
Foam sleeve 3/8" for 1/2" PEX	189 lf	leaders under other zones and at control joints
PEX staples / ties, 18" o.c.	3.3k	to the mesh or foam